LESSON 01 / 02

Status of this sheet

The equation $x^y = y^x$ and multi-base logarithmic systems are **standard IOQM/RMO algebra** and appear in specialised exam questions; everything below is derived from the Part I toolkit (monotonicity, change of base, $\frac{\ln t}{t}$) with no new theory assumed. Nothing here "can never appear in JEE" — the classification is only that these models go beyond the current syllabus sentence "Logarithms and their properties".

$x^y = y^x$ — the complete positive solution set

For $x, y > 0$: if $x = y$ everything works. If $x \neq y$, take ln: $y\ln x = x\ln y$, i.e. $\frac{\ln x}{x} = \frac{\ln y}{y}$. So x, y are two distinct points where $f(t) = \frac{\ln t}{t}$ takes the **same value** — and f increases on $(1, e]$, decreases on $[e, \infty)$, so any level is cut at most twice.

SHORT DERIVATION

Parametrisation of all unequal solutions

Put $y = tx$ with $t > 0$, $t \neq 1$. Then $x^{tx} = (tx)^x \Rightarrow x^t = tx \Rightarrow x^{t-1} = t \Rightarrow x = t^{1/(t-1)}$, $y = t^{t/(t-1)}$. Writing $t = 1 + \frac{1}{n}$ gives the classic family $x = \left(1 + \frac{1}{n}\right)^n$, $y = \left(1 + \frac{1}{n}\right)^{n+1}$ for $n > 0$ ($n \in \mathbb{Q}^+$ or $\mathbb{N}$).

Check $n = 1$: $(x, y) = (2, 4)$: $2^4 = 4^2 = 16$ ✓. Integer solutions with $x \neq y$: exactly $(2, 4)$ and $(4, 2)$.

WORKED EXAMPLE

Worked. Real solutions with $x < y$, $xy = 8$, $x^y = y^x$?

$\frac{\ln x}{x} = \frac{\ln y}{y}$ with $y = 8/x$; solve $\frac{\ln x}{x} = \frac{\ln(8/x)}{8/x}$ numerically/graphically by monotonicity arguments; or use the parametrisation: $\frac{y}{x} = t = 1 + \frac{1}{n}$, $xy = 8$ fixes n. From $y = tx$ and $xy = 8$: $tx^2 = 8$. With $x = t^{1/(t-1)}$ this becomes $t^{1 + \frac{2}{t-1}} = 8 \Rightarrow t^{\frac{t+1}{t-1}} = 8$. Trial $t = 4$: $4^{5/3} \approx 10.08 \neq 8$; $t = 2$: $2^3 = 8$ ✓. So $t = 2$, $x = 2$, $y = 4$. Answer: $(x, y) = (2, 4)$ (and the symmetric pair if order is free).

Multi-base logarithmic systems

The three moves: (1) reciprocity $\log_a b \cdot \log_b a = 1$; (2) chain $\log_a b \cdot \log_b c = \log_a c$; (3) change every base to ln (or to a common base c) and clear denominators.

WORKED EXAMPLE

Worked. Solve $\log_2 x = \log_x 4$

Domain $x > 0$, $x \neq 1$. Reciprocity: $\log_x 4 = \frac{\ln 4}{\ln x} = \frac{2\ln 2}{\ln x}$; also $\log_2 x = \frac{\ln x}{\ln 2}$. Set $u = \ln x$: $\frac{u}{\ln 2} = \frac{2\ln 2}{u} \Rightarrow u^2 = 2(\ln 2)^2 \Rightarrow u = \pm\ln 2 \cdot \sqrt{2}$. So $x = 2^{\sqrt{2}}$ or $x = 2^{-\sqrt{2}}$. Check both are > 0 and $\neq 1$ ✓. Answer: $x \in \{2^{\sqrt{2}}, 2^{-\sqrt{2}}\}$.

Towers obey a convention: $a^{b^c} = a^{(b^c)}$ — exponentiation is **right-associative**; $(a^b)^c = a^{bc} \neq a^{b^c}$ in general. Nested logs peel from the inside: $\log_2(\log_3(\log_4 256)) = \log_2(\log_3 4) = \log_2\left(\frac{\ln 4}{\ln 3}\right)$ — leave as exact expressions; never invent a "log of log" law.

COMMON TRAP

Traps on this sheet

- $x^y = y^x$ over **integers** vs **reals** give different answer sets; the parametrisation covers reals; integers need $t = 1 + 1/n$ with t rational $\Rightarrow n = 1$ only among integers (for $x < y$ positive integers: $\{2, 4\}$).
- $\log_a b = \log_a c \Rightarrow b = c$ **needs the same base**; $\log_2 x = \log_3 x \Rightarrow x = 1$ only.
- Solving $\log_x(\cdot)$ adds base conditions $x > 0$, $x \neq 1$ on top of argument positivity.
- $f(t) = \frac{\ln t}{t}$ is **not** monotone on $(0, \infty)$ — it turns at $t = e$; two solutions for $f(t) = c \in (0, 1/e)$ sit on opposite sides of e.

Assumptions & conventions

All variables positive reals unless stated; integer solutions are a separate reading of the same equations; $\frac{\ln t}{t}$ analysis uses its stated monotonicity shape (derivable later by calculus; here taken as the known shape of the standard curve, with the level-curve picture as the geometric model).

LESSON 02 / 02

Status of this sheet

Order-of-magnitude estimates and Legendre's formula are **classic IOQM/RMO number theory** built from the Part I toolkit (logs, floors, geometric progressions). The hierarchy statements below are comparisons between standard functions, models derivable from supplied information — not additional theory claimed as a syllabus item.

The growth hierarchy (state it as ordering, use it as a tool)

For fixed $a > 1$, $b > 1$, $c > 1$, $k \in \mathbb{N}$, as $x \to \infty$:

$$\log_c x \ll x \ll x^k \ll a^x \ll b^{b^x}.$$

Use in estimates: to compare $n^{\log n}$-shaped quantities, take $\log$ once or twice until everything is polynomial. Example: which is larger for huge n — 100^n or n^{100}? Take ln: $n \ln 100$ vs $100 \ln n$; divide by n: $\ln 100 \approx 4.605$ vs $\frac{100 \ln n}{n} \to 0$. So 100^n wins for large n; at $n = 100$ the two are **equal**, for $n > 100$ the exponential 100^n is larger, for $e < n < 100$ the power n^{100} is larger (crossover where $\frac{\ln n}{n} = \frac{\ln 100}{100} = 0.046$).

KEY RESULT

Digit-counting engine. $d(N) = \lfloor \lg N \rfloor + 1$ for integers $N \geq 1$. Sums of digits in base b: $N = \sum d_i b^i$ with $0 \leq d_i < b$; the digit-sum $s_b(N) = \sum d_i$ satisfies $N - s_b(N)$ is divisible by $b - 1$ (casting out $b - 1$'s). This pair ($\lfloor \log \rfloor$, digit sum) is exactly what Legendre's formula packages.

Legendre's formula — the exponent of a prime in $n!$

For prime p: $v_p(n!) = \sum_{k=1}^{\infty} \left\lfloor \frac{n}{p^k} \right\rfloor = \frac{n - s_p(n)}{p - 1}$,

where $s_p(n)$ is the digit sum of n in base p (finitely many non-zero terms: stop when $p^k > n$).

SHORT DERIVATION

Derivation — count multiples, then clean up with digits

Among $1, 2, \ldots, n$, exactly $\lfloor n/p \rfloor$ are divisible by p; among those, $\lfloor n/p^2 \rfloor$ contribute a **second** factor p; and so on — each integer m contributes $v_p(m)$ factors, so the total is $\sum_{k \geq 1} \lfloor n/p^k \rfloor$ ✓ (count each m once per power of p dividing it). For the closed form, write $n = s_p(n) + (p - 1) \sum_{k \geq 1} \lfloor n/p^k \rfloor$ (base-p identity: $\lfloor n/p^k \rfloor = \sum_{j > k} d_j p^{j-k-1}$), rearranged to the formula.

WORKED EXAMPLE

Worked. $v_2(100!)$ and $v_5(100!)$
$v_2(100!) = 50 + 25 + 12 + 6 + 3 + 1 = 97$. *Via digits:* $100 = (1100100)_2$, $s_2 = 3$, $\frac{100-3}{1} = 97$ ✓. $v_5(100!) = 20 + 4 = 24$; *digits:* $100 = (400)_5$, $s_5 = 4$, $\frac{100-4}{4} = 24$ ✓. *Consequence:* $100!$ *ends in exactly* 24 *zeros (the power of* $10 = 2 \cdot 5$ *is the smaller of* v_2, v_5*).*

Series estimates with logs and floors

Truncation errors: $\sum_{k=1}^{n} \frac{1}{k}$ *grows like* $\ln n$ *(harmonic estimate:* $\ln(n+1) \le H_n \le 1 + \ln n$ *— provable by comparing the sum with* $\int \frac{dx}{x}$*, a standard derived model). The telescoping estimate* $\frac{1}{k+1} \le \ln\left(1 + \frac{1}{k}\right) \le \frac{1}{k}$ *(from* $\frac{1}{1+t} \le \ln(1+t)$*-style log bounds at* $t = 1/k$*) is the workhorse behind it.*

WORKED EXAMPLE

Worked. Bound $H_{100} = 1 + \frac{1}{2} + \cdots + \frac{1}{100}$
$\ln 101 \approx 4.615 \le H_{100} \le 1 + \ln 100 \approx 5.605$ *; the true value is* ≈ 5.187 ✓ *inside the bracket. Use such brackets to decide sign/size questions without computing sums.*

COMMON TRAP

Traps on this sheet

- $\lfloor n/p \rfloor + \lfloor n/p^2 \rfloor + \cdots$ *is* ***not*** *a geometric series — do not sum it as one; each term is a floor.*
- *Trailing zeros of* $n!$ *are* $v_5(n!)$ *(5s are rarer than 2s), not* $v_2(n!)$*.*
- log *estimates compare growth rates; a bound valid "for large* n*" must not be quoted for a specific small* n *without checking (e.g.* $n^{100} > 100^n$ *fails for* $n = 2$*).*
- *Digit sums* s_p *depend on the base;* $s_2(100) = 3$ *counts* ***ones in binary****, not decimal digits.*

Assumptions & conventions

p prime; $v_p(m)$ = exact exponent of p in m; floors are real floors; harmonic estimates quoted as standard derived bounds (checked numerically above); hierarchy statements are eventual comparisons (there is always a crossover point).